

Outset · AI in Healthcare · Day 3 · Group 6

Does estrogen keep older women's minds sharp?

A causal question, where a naive analysis gets it wrong -- and the right method flips the answer.

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Outset

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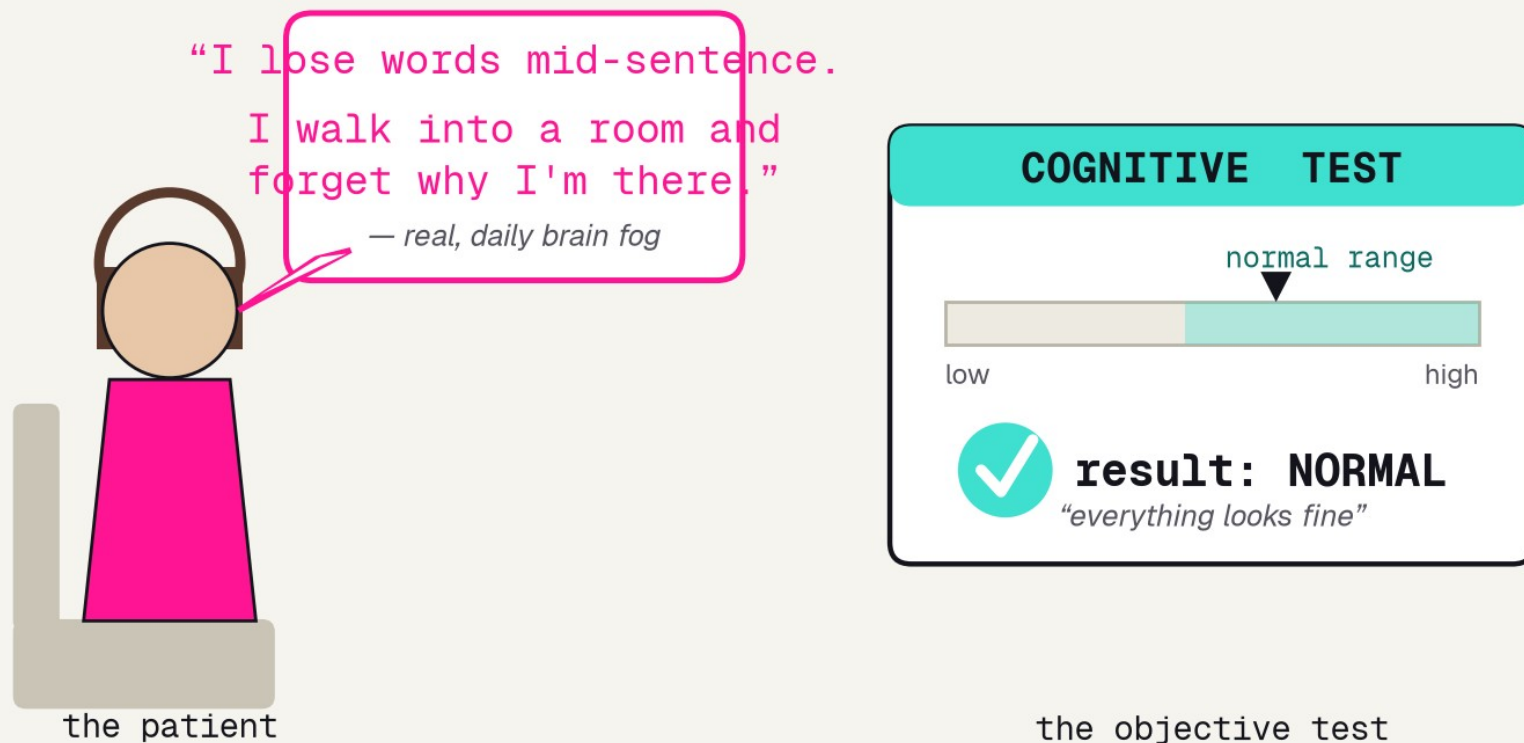
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BACKGROUND

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The visit no test explains

In the exam room: a symptom the test can't see



concept -- Maki 2024 (menopause & brain fog) / Chen & Shafir 2025 (dismissed symptoms), Harvard Health

She describes real, daily brain fog. The objective test says "normal." Both can be true -- and the gap is the whole project.

The tests say fine; she says not fine

THE MISMATCH

Objective scores mostly land in the normal band, yet self-reported fog is high and scattered. The lines cross.

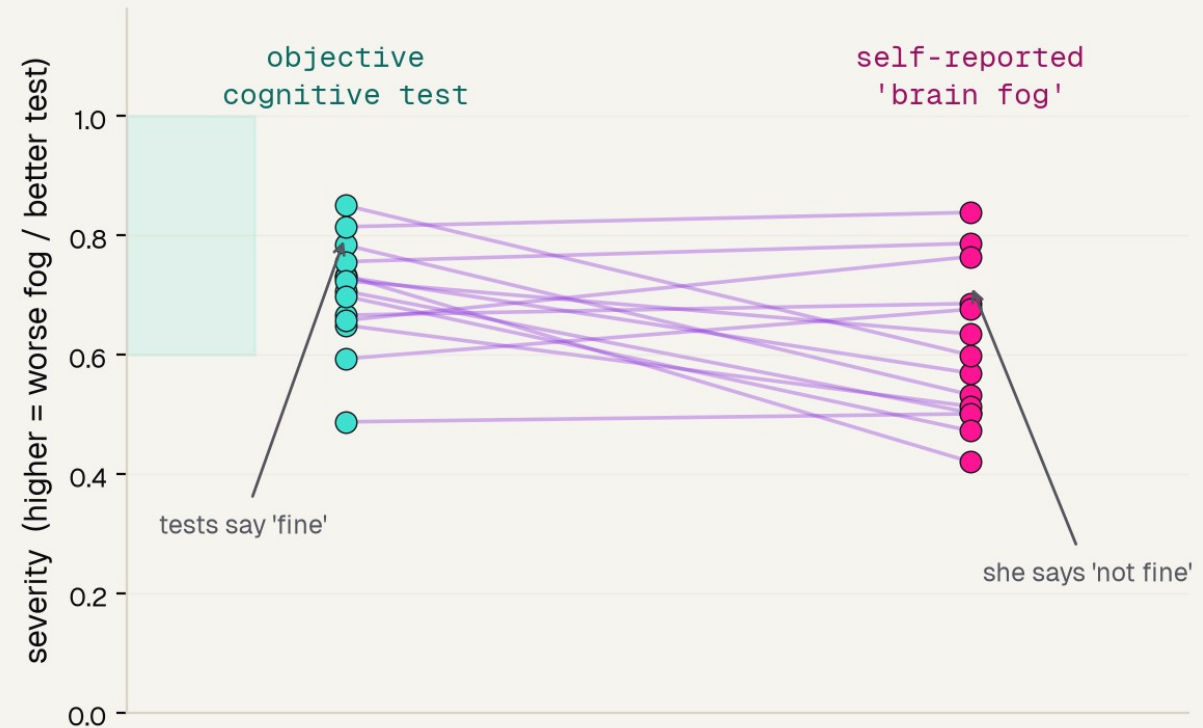
WHY IT MATTERS

A test that reads "fine" can miss a symptom a woman lives with every day.

THE STANCE

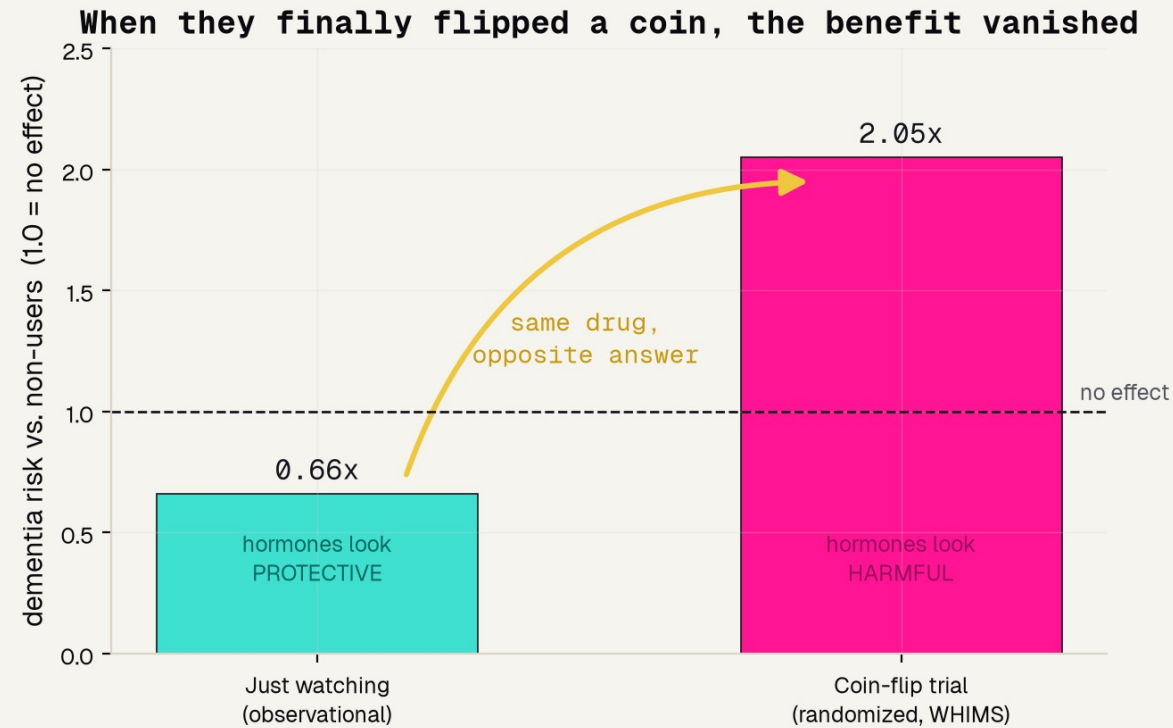
Take both as real evidence: the number AND the account. Neither is noise.

The tests can say 'normal' while the patient feels foggy



adapted from Maki 2024, Harvard Health (menopause & brain fog)

Twenty years of watching said estrogen helps



adapted from Tang 1996 / Zandi 2002 (observational) vs. Shumaker 2003 WHIMS (randomized)

0.66x

the risk observational studies saw -- looks protective

2.05x

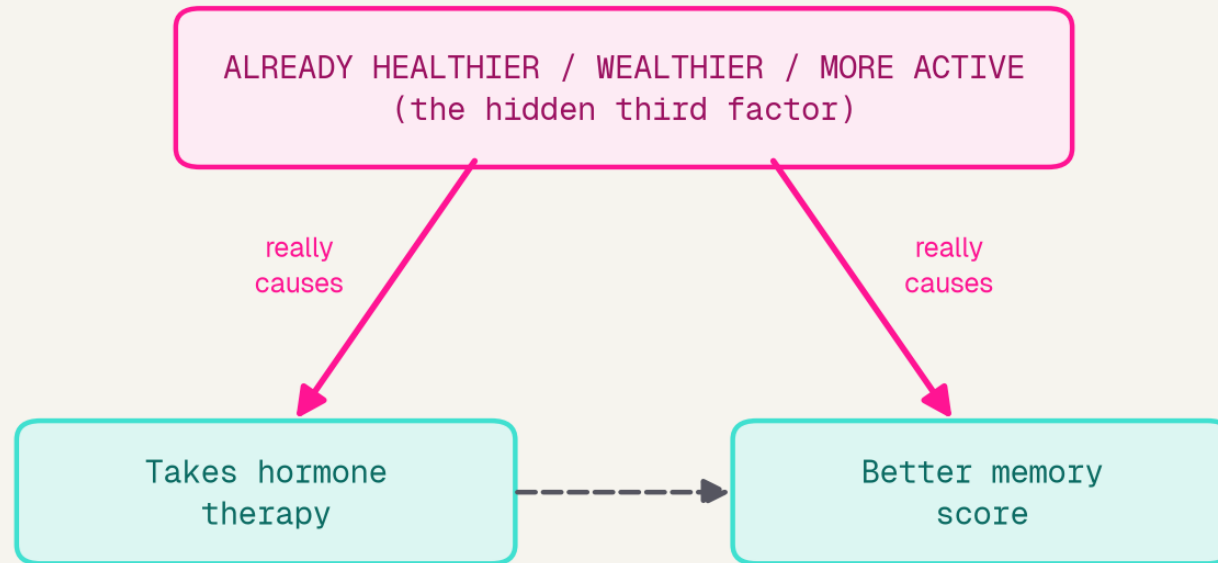
the risk the coin-flip trial found -- actual harm

2003

the year WHIMS overturned twenty years of headlines

Why watching lied: healthy-user bias

Healthy-user bias: a hidden factor fakes the link



looks like cause -- but isn't

concept -- adapted from Wharton 2009 / Vandenbroucke 2009

WHO CHOSE THE PILL

Users were richer, more active, and saw doctors more often to begin with.

THE HIDDEN FACTOR

That same head start drives BOTH the drug use AND the better memory.

SO THE DIRECT LINK

only looks like cause. It is a mirage -- confounding, a.k.a. healthy-user bias.

THE DATA

Why NHANES

739 WOMEN

all age 60+, drawn from the NHANES 2013-14 US national health survey

USED ESTROGEN?

whether she ever took female hormones -- the treatment we are studying

HER BACKGROUND

age, education, and family income -- the confounders we must adjust for

MEMORY SCORE

a digit-symbol test; higher is sharper. We study who lands in the weaker half

The honest question we can actually ask

We cannot flip a coin to assign estrogen in a survey -- so we cannot prove cause. Instead we ask a narrower, honest question:

Does the estrogen effect SURVIVE once we account for age, education, and income?

CRUDE

Estrogen's effect all by itself -- ignoring everything else about the woman.

ADJUSTED

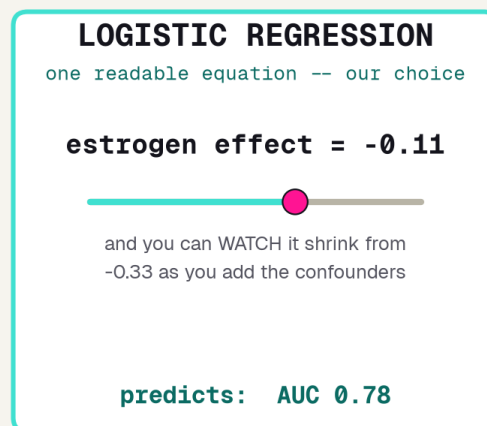
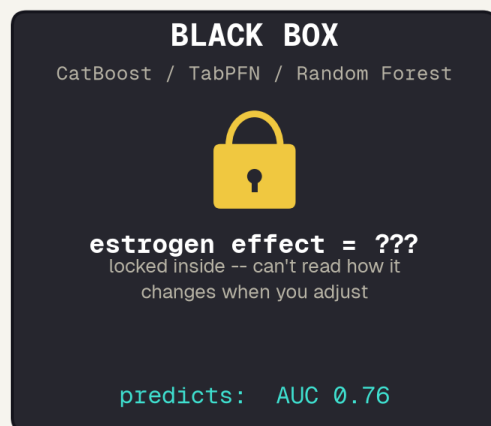
Estrogen's effect AFTER accounting for age, education, and income.

If the effect shrinks from crude to adjusted, the headline was mostly confounding -- not cause.

THE MODEL

Prediction is not causation

Prediction is not causation: which model can answer the question?

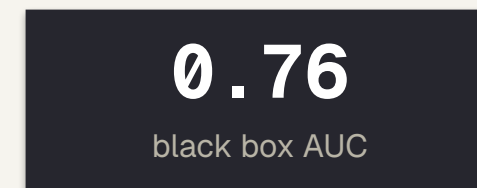
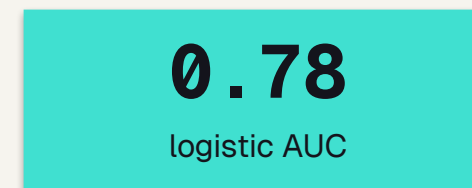


A causal question needs a model you can READ -- and here it predicts just as well anyway.

AUCs = 5-fold cross-validation on our NHANES sample (illustrative model-choice figure)

A CAUSAL QUESTION FORCES THE MODEL

To answer it you must READ the estrogen coefficient and watch it change as you adjust. Only an interpretable model gives you that number.



The black box does not even predict better here -- and it can never hand you the effect size. So we choose logistic regression.

Why logistic regression, and the exact recipe

1

MODEL

LogisticRegression -- one linear equation whose estrogen coefficient is the effect size

2

FEATURES

age, education, income, used_estrogen, years_estrogen -- each standardized so coefficients compare

3

THE CAUSAL MOVE

fit the estrogen effect TWICE: crude (alone) then adjusted (+ confounders); the change is the confounding

4

PREDICTIVE SCORE

AUC via 5-fold cross-validation, so no woman is graded by a model that trained on her

5

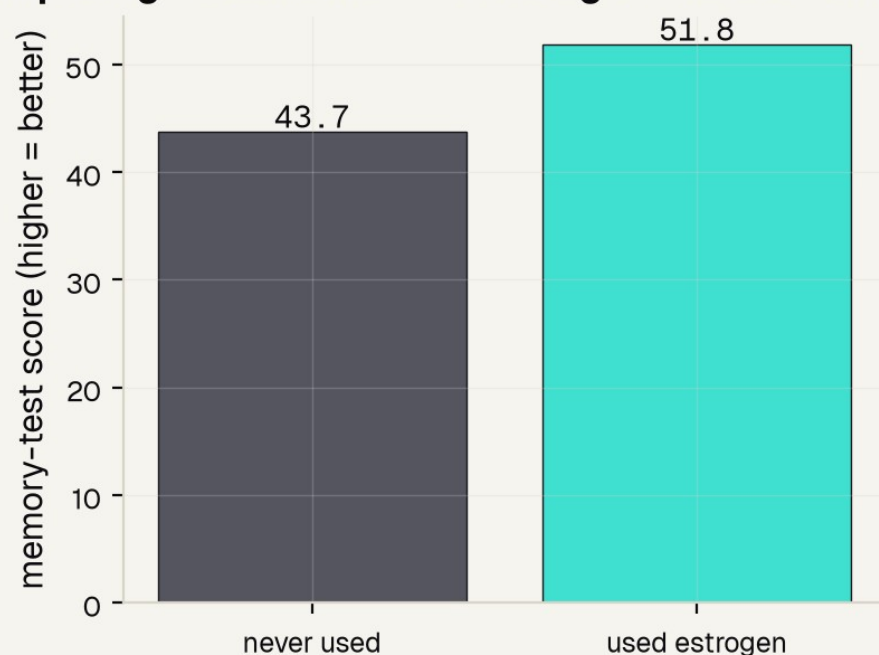
FAIRNESS

a held-out train/test split, with accuracy reported separately for each group

THE RESULTS

The tempting headline

The tempting headline: 'estrogen users think better'



+8.1

points -- the raw gap, users over non-users

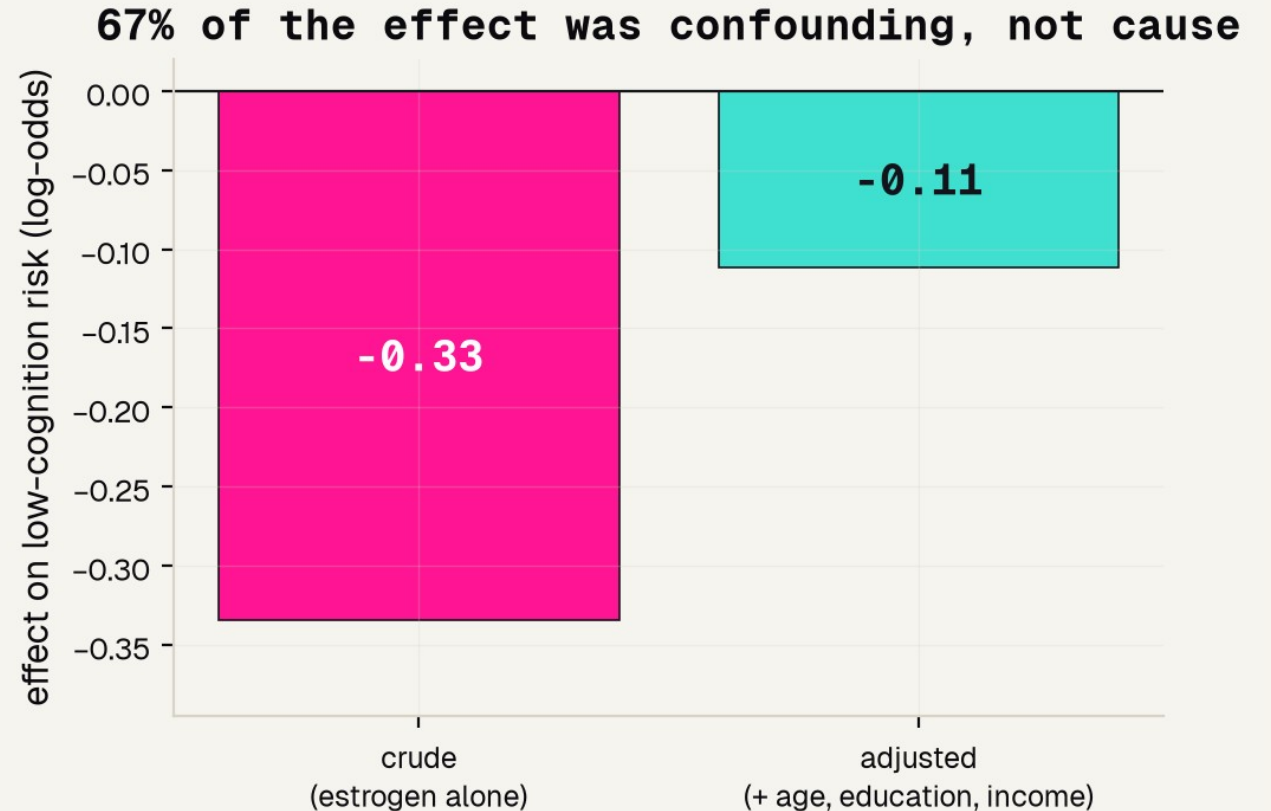
51.8 versus 43.7. Taken at face value, the exciting headline writes itself: "estrogen users think better." It is the same clue the 1990s studies chased -- but we never checked whether the two groups were the same kind of people to begin with.

Most of the gap was confounding

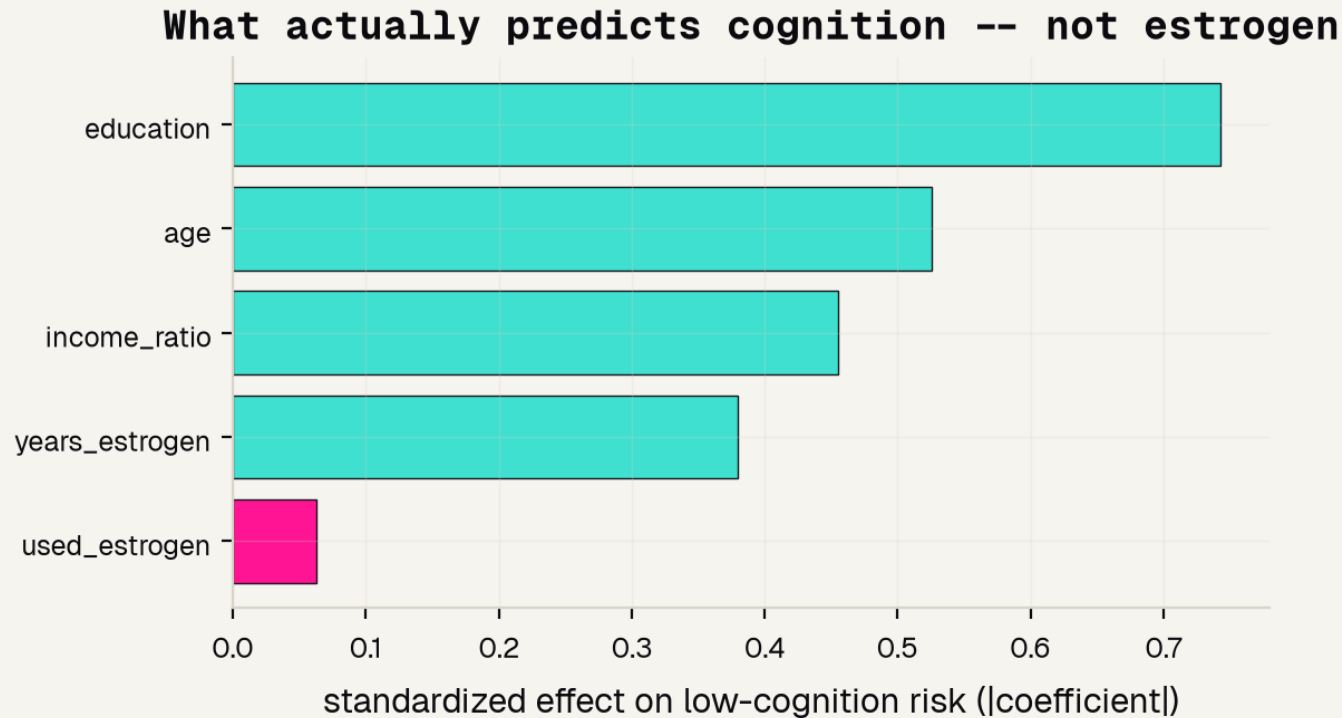
67%

of the apparent effect was confounding

The crude effect (-0.33) shrank to just -0.11 once we adjusted for age, education, and income. That missing two-thirds was never the estrogen -- it was that users were healthier and better-off to start with.



What actually predicts cognition



FROM THE SAME MODEL

education the biggest driver

age close behind

income still matters

estrogen the smallest bar

Feature importance is free from an interpretable model: it is the standardized coefficients. Prediction and causation agree, and neither points at the pill.

Can it even predict?

0.78

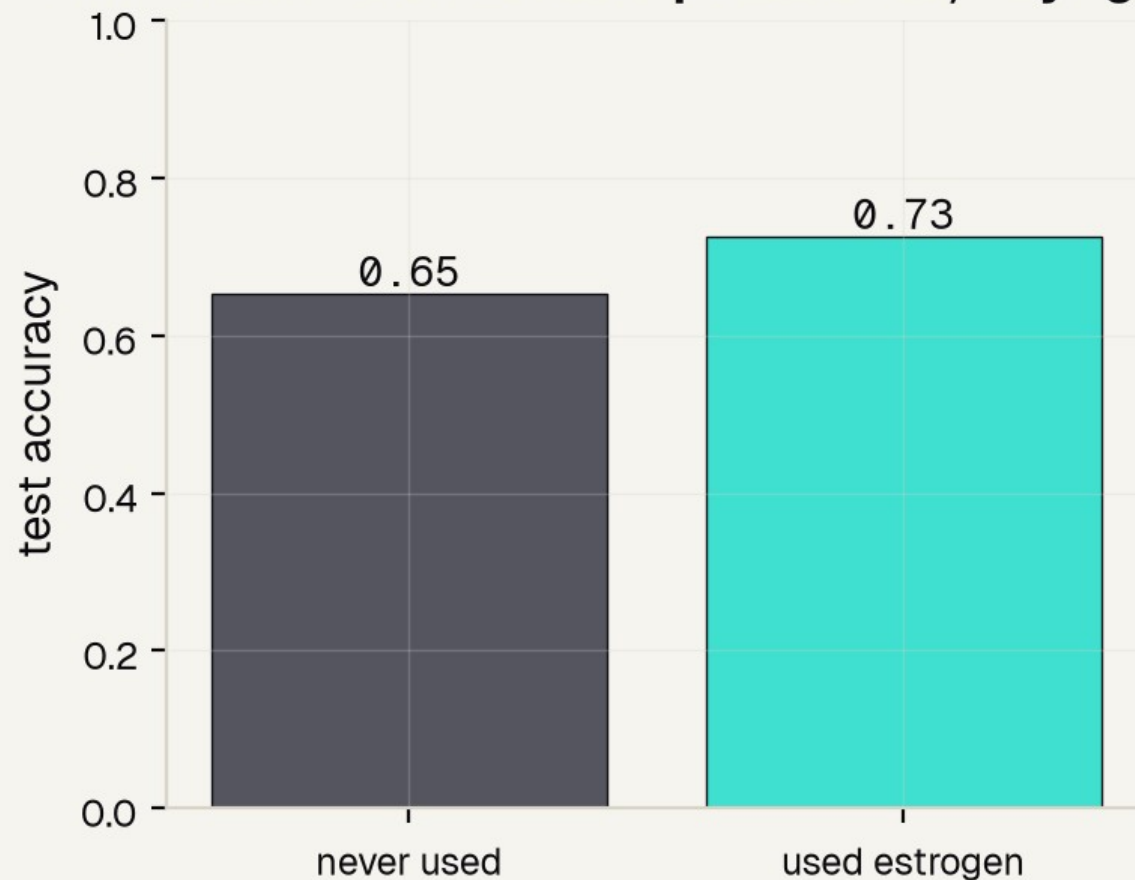
predictive AUC

5-fold cross-validation. 0.5 would be a coin flip, so 0.78 is a real signal -- but it is driven by age and education, not estrogen.

An honest secondary number -- and a reminder: a black box scored 0.76 here, no better, and could give us no effect size at all.

Does it work for both groups?

How well the model predicts, by group



0.73

accuracy for estrogen-USERS

0.65

accuracy for NON-users

A tool reliably right for one group and shakier for another can quietly do harm if a clinic trusts it equally. Naming the gap is the honest move.

THE TAKEAWAY

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What a survey can and cannot do

WHAT IT CAN DO

Show a tempting headline is mostly a mirage, and that adjusting for background shrinks the effect by 67%.

the mirage, exposed

WHAT IT CANNOT DO

Prove cause. We only corrected for confounders we measured and thought of; something unrecorded could still be doing the work.

a hint, not a proof

WHAT SETTLES IT

A randomized trial: a coin flip, not the patient, assigns the drug, so the groups start the same. That is why WHIMS could overturn 20 years of headlines.

only an experiment

References

OBSERVATIONAL & TRIAL

- [1] Tang et al. 1996, Lancet -- observational; users got Alzheimer's less/later.
- [2] Zandi et al. 2002, JAMA (Cache County) -- observational; long-term users lower risk.
- [3] Shumaker et al. 2003, JAMA -- WHIMS RCT; hormones DOUBLED dementia risk in 65+.
- [4] Rapp et al. 2003, JAMA -- WHIMS RCT; no cognitive benefit, scores trended worse.

BIAS & SYMPTOMS

- [5] Wharton et al. 2009, Maturitas -- healthy-user bias; users start out healthier.
- [6] Vandenbroucke 2009, Lancet -- why observational studies and RCTs disagreed.
- [7] Maki 2024, Harvard Health -- menopause brain fog vs. normal test scores.
- [8] Chen & Shafir 2025, Harvard Health -- the dismissal of women's symptoms.

The one habit to carry with you

Suspect the chooser, not the treatment.

When a group that CHOSE a treatment looks healthier, the head start usually belongs to the person, not the pill -- and only a coin-flip experiment can promote a correlation to a cause.

And take the patient's account seriously -- even when the test says "fine."

Thank You

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